# Electrolysis

## Move charged particles

### 1 Electrolyte & electrodes

Ions move only when molten/dissolved. Positive ions go to negative cathode; negative ions to positive anode.

Cathode: gain electrons, reduction. Anode: lose electrons, oxidation. Electrons move in wires, ions in liquid.

### 2 Molten binary compounds

Metal at cathode; non-metal at anode. No competing water ions.

Molten $PbBr_2$: $Pb^{2+} + 2e^- \rightarrow Pb$; $2Br^- \rightarrow Br_2 + 2e^-$.

Balance atoms and total charge; halogens are diatomic.

## Predict the products

### 3 Aqueous product rules

Water adds $H^+$ and $OH^-$. At cathode: $H_2$ if metal is above hydrogen, otherwise metal.

At anode with inert electrodes: halogen if halide present, otherwise $O_2$.

NaCl(aq): $H_2/Cl_2$; $CuCl_2$(aq): $Cu/Cl_2$; $CuSO_4$(aq): $Cu/O_2$.

### 4 Useful half-equations

$2H^+ + 2e^- \rightarrow H_2$. $Cu^{2+} + 2e^- \rightarrow Cu$.

$2Cl^- \rightarrow Cl_2 + 2e^-$. $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$.

Oxidation puts electrons on product side; reduction uses them on reactant side.

## Apply & investigate

### 5 Aluminium extraction

Molten $Al_2O_3$/cryolite mixture needs lower temperature than pure $Al_2O_3$. $Al^{3+} + 3e^- \rightarrow Al$. $2O^{2-} \rightarrow O_2 + 4e^-$. Oxygen reacts with carbon anodes → $CO_2$; replace anodes.

Energy needed for heat and current; carbon cannot reduce aluminium oxide.

### 6 Aqueous practical

Hypothesis: changing dissolved compound changes products. Use inert electrodes/DC; compare solutions at same voltage, time, concentration and electrode setup.

Test $H_2$: pop; $O_2$: relights glowing splint; $Cl_2$: bleaches damp litmus. Record metal deposits too.

Repeat; goggles, ventilation for chlorine; switch off before adjusting electrodes.